Impact audit: the pre-2026-10-18 `rho.quartic` / `rho.cubic` class set-up undercharge

2026-10-24 · `research/class_setup_impact_20261024/` · derived from committed records with ecbench `compare` and `table`; nothing re-measured, no committed session changed · class: **accounting** (no method improves)

What was wrong

`rho.quartic` and `rho.cubic` begin with a class set-up: the scalar multiplication that fixes ι (j = 1728) or ζ_3 (j = 0) on the subgroup. Binaries built before `9ead78cf0` (2026-10-18) charged it to the `setup` phase but not to `total_gae` or `S`. Each total fell short of the phase sum by exactly the set-up: 16–58 GAE at r ≈ 2^14–2^24, and 111–125 GAE at r ≈ 2^39.5. The phase sum is the true total. Every other arm, `rho.negation` included, was charged correctly.

`ecbench verify` reconstructs the old figure from a replay so that old records still replay exactly. It does this only for records from the five pre-fix producer binaries, with no `detail.class_setup`, on these two methods. A record the current harness writes, undercharged and resealed, fails its replay (`docs/ecbench/CLASS_SETUP_AUDIT.md`).

Census (every session, every storage form, four repositories)

study	sessions	records	binary	commit
prime_rho_automorphism_20261015	2	192	`0a8a48d7…`	`c9db37d01`
drastic_prime_conductor_gap_round2	2	128	`1a87842e…`	`466035c18`
drastic_prime_conductor_gap_round3	3	320	`d24c2247…`	`7fa17f92e`
volcano_ecdlp_census_20261017 (V3, `.tar.xz`)	3	4,032	`ac4bc5b2…`	`d13f5259c`
volcano_ecdlp_census_followup_20261023 (V5, `.tar.xz`)	66	7,600	`ac4bc5b2…`	`d13f5259c`
isogeny-volcano-cost_20261017 (gzip)	2	256	`7da27ec3…`	`3d191d806`

Recorded against corrected

Both columns come from the same `ecbench compare`: operations, ratio of mean S, linearised-t 95% interval. One is run on the session as recorded, the other on a scratch copy whose totals include the set-up.

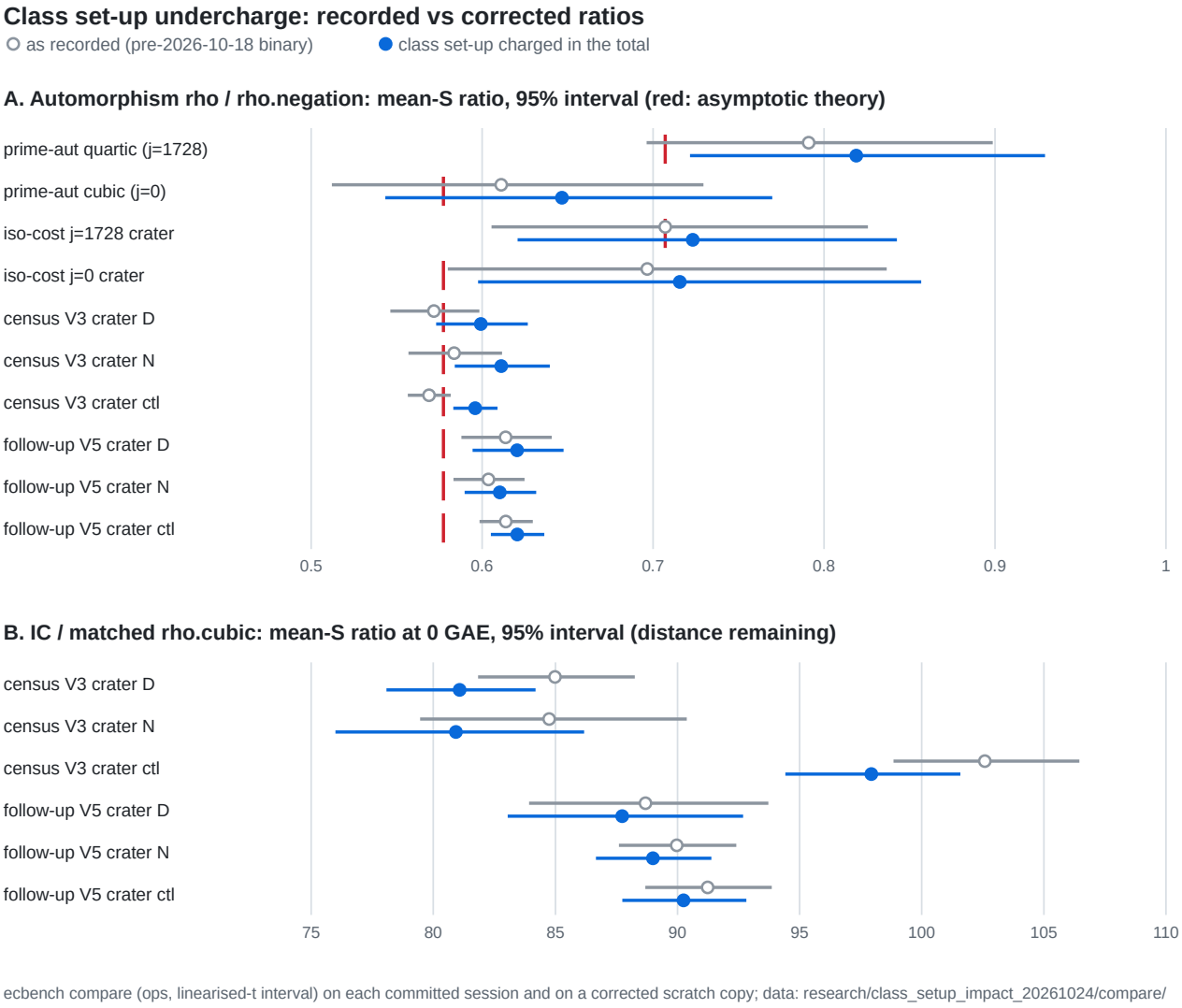


Figure 1: Recorded (open) and corrected (filled) ratios. Red ticks: asymptotic theory. Data: `compare/*.json`.

automorphism rho / rho.negation	pairs	recorded	corrected	theory
prime-aut quartic	72	0.791 [0.696, 0.899]	**0.819 [0.722, 0.929]**	0.707
prime-aut cubic	72	0.611 [0.512, 0.729]	**0.647 [0.543, 0.770]**	0.577
iso-cost j = 0 crater	64	0.697 [0.580, 0.837]	**0.716 [0.598, 0.857]**	0.577
iso-cost j = 1728 crater	64	0.707 [0.606, 0.826]	**0.723 [0.621, 0.843]**	0.707
census V3 crater D	1,000	0.572 [0.546, 0.598]	**0.599 [0.573, 0.627]**	0.577
census V3 crater N	1,000	0.584 [0.557, 0.612]	**0.611 [0.584, 0.640]**	0.577
census V3 crater ctl	2,000	0.569 [0.557, 0.582]	**0.596 [0.583, 0.609]**	0.577
follow-up V5 crater D	1,000	0.614 [0.588, 0.641]	**0.620 [0.594, 0.648]**	0.577
follow-up V5 crater N	1,000	0.604 [0.583, 0.625]	**0.610 [0.590, 0.632]**	0.577
follow-up V5 crater ctl	2,000	0.614 [0.598, 0.630]	**0.621 [0.605, 0.636]**	0.577

IC / matched rho.cubic, 0 GAE	pairs	recorded	corrected
census V3 crater D	1,000	85.0× [81.8, 88.3]	**81.1× [78.1, 84.2]**
census V3 crater N	1,000	84.7× [79.5, 90.4]	**80.9× [76.0, 86.2]**
census V3 crater ctl	2,000	102.6× [98.8, 106.5]	**97.9× [94.4, 101.6]**
follow-up V5 crater D	1,000	88.7× [83.9, 93.7]	**87.7× [83.0, 92.7]**
follow-up V5 crater N	1,000	90.0× [87.6, 92.4]	**89.0× [86.7, 91.4]**
follow-up V5 crater ctl	2,000	91.2× [88.7, 93.9]	**90.2× [87.7, 92.8]**

The correction raises only the rho arm's charged cost, so IC/rho falls by the same factor at 0, 0.1 and 1 GAE. Rho stays ahead on every row, by about 81× or more. This is accounting, not an IC improvement, and no IC baseline moves. The 40-bit conductor-gap rows move by 0.012–0.026% (`tables/`), which is invisible at the precision they report.

What changes

1. **prime_rho_automorphism_20261015.** The quartic control now excludes its prediction 0.707, so finding 1 holds for cubic only. The set-up is a fixed cost of about one scalar multiplication, and the asymptotic √ saving leaves it out. The round-2 factors become about 1.55× and 1.22×, and round 2′s conclusion stands.
2. **volcano_ecdlp_census_20261017.** The positive control still passes its gate (upper bound below 0.85). It sits 3–6% above 1/√3, and the N and ctl intervals exclude 0.577. The crater IC / `rho-cubic` gap is about 81×, not 85×. Erratum E-CS in `REPORT.md` supersedes 0.5733 and 84.9×, including in the validated `benchmarks/README.md` row, which is digest-pinned and not rewritten here.
3. **volcano_ecdlp_census_followup_20261023 (in progress).** It runs the same frozen binary, so its preregistered analysis will read `rho-cubic` about 1% low: the positive control, the crater gap and the H-T composition. It should add the set-up to every `rho-cubic` total, or amend its plan.
4. **isogeny-volcano-cost_20261017.** Both controls remain detected. j = 1728 still covers √(1/2).
5. **conductor-gap rounds 2 and 3.** Checked and unchanged.